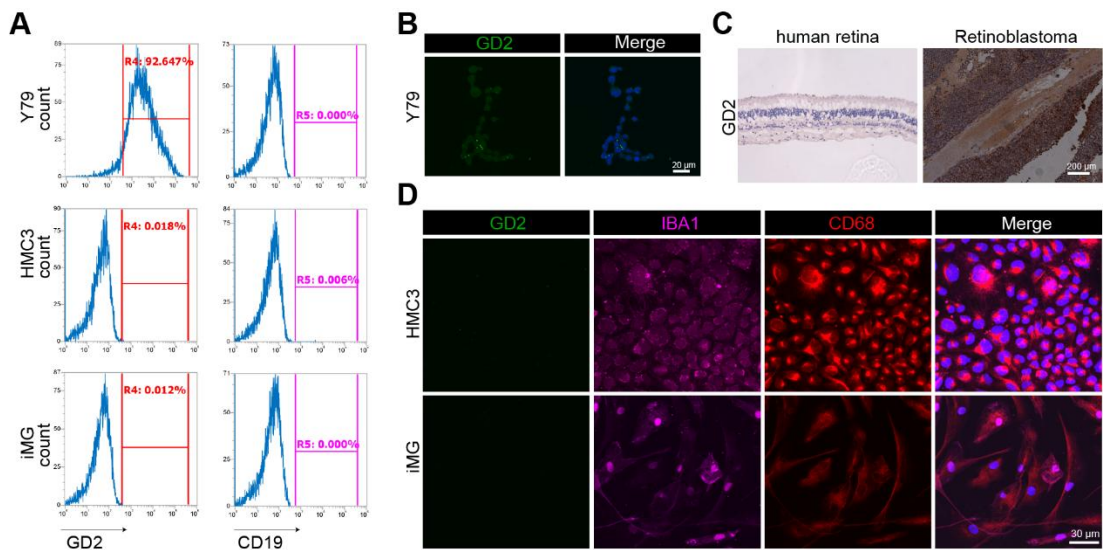
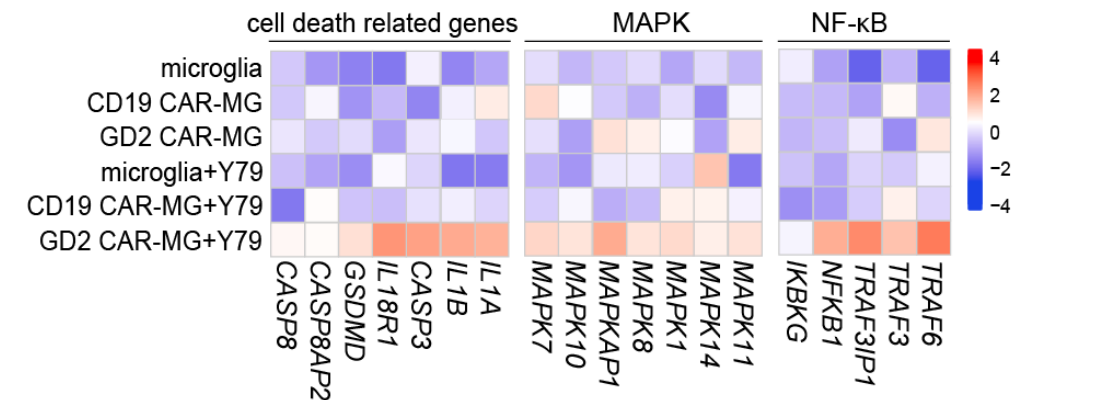


Supplementary Material

Supplementary Figures and legends

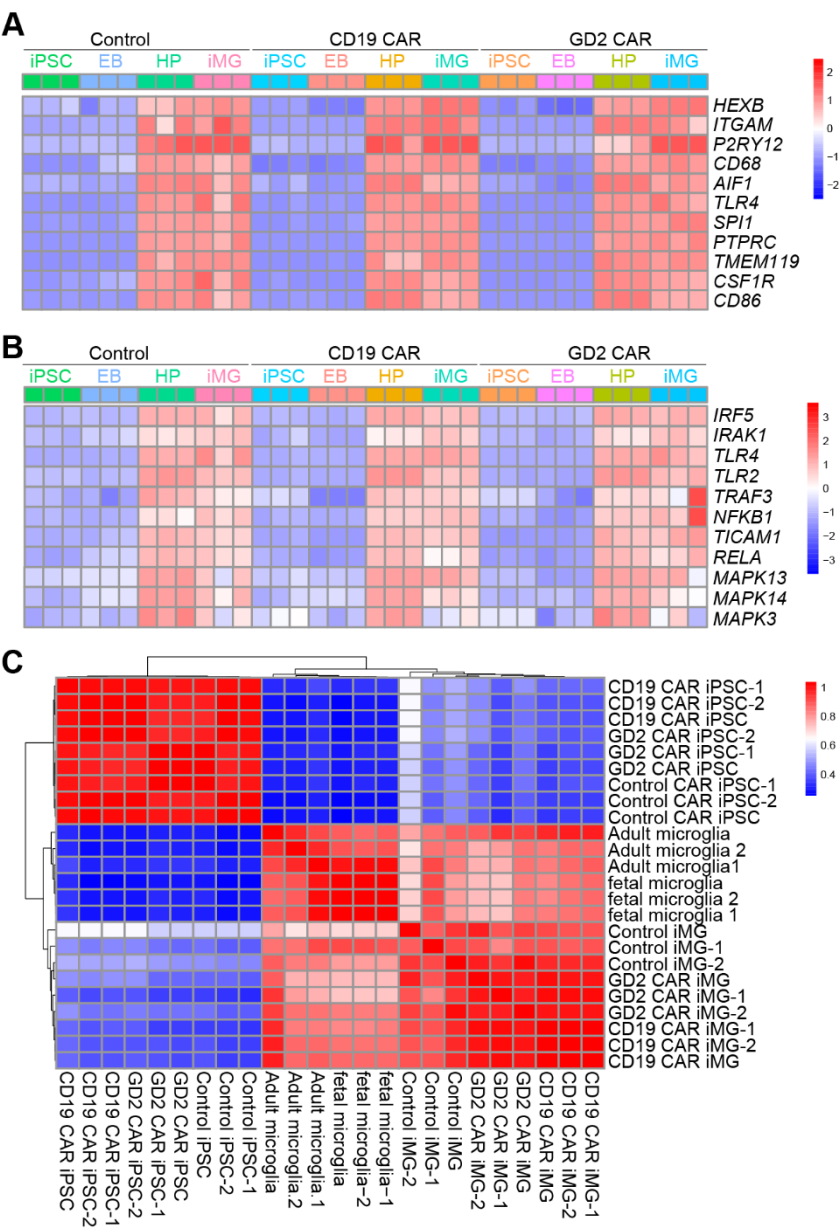


Supplementary Figure S1. GD2 is expressed in retinoblastoma. (A) FACS analysis of GD2 and CD19 expression in Y79 retinoblastoma cell line, HMC3 microglia cell line and iMG cells. (B) Representative images of Y79 immunostained for GD2 (Green). Cell nuclei were stained with DAPI (blue). Scale bar, 20 μ m. (C) Immunohistochemistry analysis of GD2 expression in human retina and retinoblastoma. Scale bar, 200 μ m. (D) Representative images of microglia cell line HMC3 and iMG cells immunostained for GD2 (green), microglia marker IBA1 (rose red) and CD68 (red). Cell nuclei were stained with DAPI (blue). Scale bar, 30 μ m.

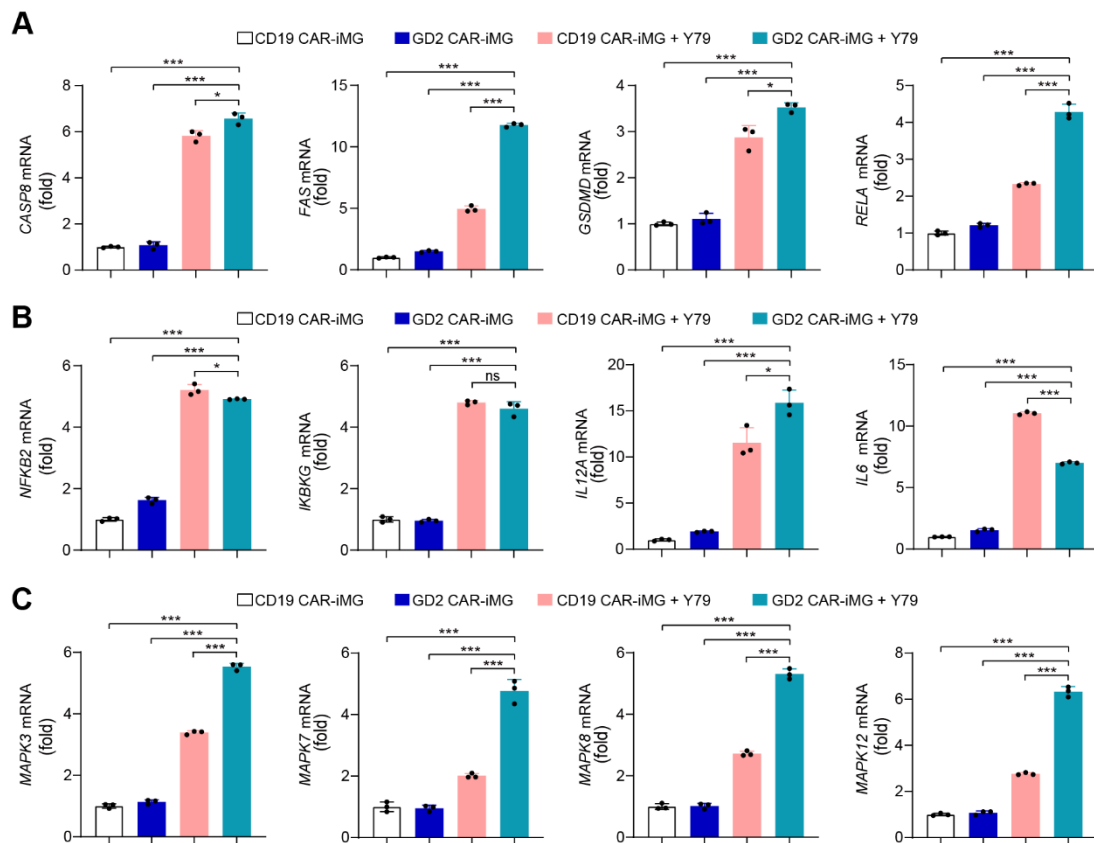


Supplementary Figure S2. Cell death related genes and microglia activated related

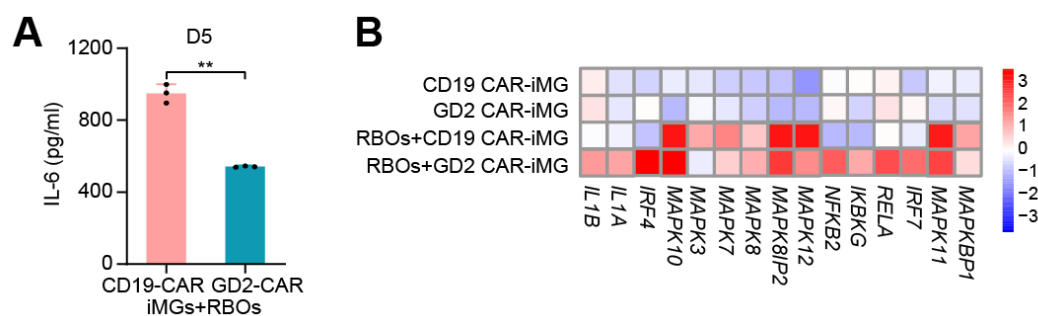
MAPK and NF- κ B signaling pathway genes were upregulated in GD2 CAR-microglia co-cultured Y79. MG means microglia.



Supplementary Figure S3. Generation of GD2 CAR-iMG using iPSC. (A) Heatmap analysis of microglia marker genes expression during the differentiation of CAR-iMG derived from iPSCs. (B) Heatmap analysis of microglia activation related genes expression during the differentiation of CAR-iMG derived from iPSCs. (C) The CAR-iMGs transcriptome was profiled in comparison to human fetal microglia and adult microglia from previous studies (GSE89189).



Supplementary Figure S4. Quantitative real-time PCR analysis of microglia activated related cytokines, MAPK and NF- κ B signaling pathway related genes expression in Y79 tumor cells co-cultured CAR-iMG. (A) The qRT-PCR analysis of *CASP8*, *FAS*, *GSDMD* and *RELA* expression in CAR-iMG with or without Y79 tumor cells. (B) The qRT-PCR analysis of *NFKB2*, *IKBKG*, *IL12A* and *IL6* expression in CAR-iMG with or without Y79 tumor cells. (C) The qRT-PCR analysis of *MAPK3*, *MAPK7*, *MAPK8* and *MAPK12* expression in CAR-iMG with or without Y79 tumor cells.



Supplementary Figure S5. Anti-tumor effects of GD2 CAR-iMG derived from iPSC in organoids. (A) ELISA analysis of IL-6 protein level in the supernatant of CAR-iMG and RBOs cocultured wells for 5 days (D5). (B) Microglial activated related cytokines, MAPK and NF- κ B signaling pathway related genes were upregulated in tumor cells co-cultured CAR-iMG.

Supplementary Table S1. Materials resources table.

Material	Company	Catalog Number	Origin
DMEM basic	Gibco	C11965500BT	Suzhou, China
DMEM/F12	Gibco	C11330500BT	Suzhou, China
B27	Gibco	17504-044	Grand Island, USA
Dispase (1U/mL)	STEMCELL Technologies	07923	Vancouver, Canada
Acctuase	STEMCELL Technologies	07920	Vancouver, Canada
DMEM/F12	Gibco	C11330500BT	Suzhou, China
DPBS	Gibco	C141905005BT	Suzhou, China
EDTA	Thermo	15575020	Vilnius, Lithuania
F12	Gibco	C11765500BT	Suzhou, China
Fetal Bovine Serum (FBS)	VivaCell	C04002-500	Uruguay
TeSR-E8 medium	STEMCELL Technologies	05990	Vancouver, Canada
Gelatin	Sigma-aldrich	G7041-100G	Vancouver, Canada

Glutamax	Gibco	35050-061	Grand Island, USA
IL-3	R&D Systems	203-IL-050	Minneapolis, USA
IL-34	PeproTech	200-34-50UG	Suzhou, China
KSR	Gibco	10828028	Grand Island, USA
M-CSF	R&D Systems	216-MC-500	Minneapolis, USA
MEM Non-essential Amino Acid Solution	Sigma-aldrich	M7145	Vancouver, Canada
N2	Life Technologies	17502-048	Grand Island, USA
Neurobasal	Gibco	21103-049	Grand Island, USA
Pen/strep	Gibco	15140-122	Grand Island, USA
Matrigel	Corning	356231	Corning, USA
Taurine	Sigma-aldrich	T-8691-25G	Vancouver, Canada
X-ViVO	LONZA	04-418Q	Walkersville, USA
Y27632	Selleck	S1049	Houston, USA
β-mercaptoethanol	Life Technologies	21985-023	Grand Island, USA
PRMI1640 medium	Gibco	A1049101	Grand Island, USA
Trizol	ThermoFisher	15596-026	Carlsbad, USA
ELISA MAX™ Deluxe Set Human IL-6	Biolegend	430504	San Diego, USA
ELISA MAX™ Deluxe Set Human TNF-α	Biolegend	430204	San Diego, USA
Bright-Lite Luciferase assay system	Vazyme	DD1204	Nanjing, China
anti-GD2	BD	554272	San Diego, USA

anti-CX3CR1	Abcam	ab8021	Cambridge, UK
anti-IBA1	Abcam	ab5076	Cambridge, UK
anti-CD68	Santa Cruz Biotechnology	sc-70761	Santa Cruz, USA
G4S linker Alexa Fluor 647	Cell Signaling technology	69782S	Danvers, USA
anti-mouse/human CD11b Pacific Blue	Biolegend	101223	San Diego, USA
anti-human CD45 Brilliant Violet 605	Biolegend	368523	San Diego, USA
anti-human GD2 APC	Biolegend	357305	San Diego, USA
anti-human CD19 Pacific Blue	Biolegend	302232	San Diego, USA